

Multi-Scale Features Fusion Convolutional Neural Networks for Rice Leaf Disease Identification

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Abstract. It is time-consuming and labor-intensive to detect rice diseases manually. The purpose of this research is to develop a convolutional neural networks (CNNs)-based system to automatically detect the diseased rice leaf infected with rice leaf blast, helminthosporium leaf blight, and bacterial leaf blight. The sizes of rice leaf spots vary with the severity of disease infection. A single model based CNN cannot effectively classify images, especially for images with objects of small size as well as multiple object scales, and complicated image background. In this research, a multiscale serial convolution neural network (MSSCNN) and a multiscale parallel convolution neural network (MSPCNN) are proposed to identify diseased rice leaf spots based on multi-modal fusion to extract different perception features and combine them to improve the performances obtained by using only one modality. Experimental results delineate that MSSCNN and MSPCNN can get better performance in identifying the diseased rice leaves. In MSPCNN, the features of tiny spots on diseased rice leaves can be completely preserved. MSPCNN can hence offer better performances than MSSCNN. Additionally, MSPCNN architecture is suitable for parallel computing environment. © 2022 Society for Imaging Science and Technology.
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1. INTRODUCTION

Kumar & Ashraf [1] opined that the rice leaf blast is estimated to cause the rice yield loss by 10%–30%. According to the survey of related literature [2, 3], the helminthosporium leaf blight is estimated to cause severe rice yield loss, and the bacterial leaf blight to cause the rice yield loss by 20%–30%,

or even up to 50%. It is undeniable that these diseases cause great harm in rice production. Developing an automated rice disease detection system is therefore necessary to increase rice yield and reduce labor cost by reducing pesticide use. The purpose of this research is to develop a rice disease diagnosis system to automatically identify the three common rice diseases: rice leaf blast, helminthosporium leaf blight, and bacterial leaf blight, so as to increase the overall rice production.

A traditional image classifier needs to extract the “Handcrafted” features [4] from the image, and then input the features into the classifier [5, 6]. However, selecting appropriate features and the right classifier usually requires some prior knowledge. CNNs can extract the features directly through the convolution kernels in the training stage, and then pass these features to the fully connected layers for classification. The feature extraction in the training stage can be performed directly by one network model. Specifically, CNNs do not require going through the process of feature and classifier selection for feasible tasks, as compared to the “Handcrafted” feature extraction method. Compared to the traditional image classification methods, several research studies [7, 8] have proved that CNNs can achieve better classification results. Generally, DCNNs (CNNs with deep layers) can learn more complex features [9]. However, it would cause an increase in computation complexity, which in turn may cause the gradient vanishing or exploding problems.

Over the past several years, artificial intelligence technology has been widely used for the diagnosis of agricultural

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diseases [10–15]. However, identifying the diseased rice leaf spots is a challenging task as the size of rice leaf is much smaller than those of general plants. To improve the identification performance, in this research, a multiscale serial convolution neural network (MSSCNN) and a multiscale parallel convolution neural network (MSPCNN) are proposed to identify rice leaf spots. Generally speaking, CNNs with deeper layers can learn more complex functions than the ones with less deep layers; thereby a CNN with deeper layers can get better results than the one with shallower layers. Therefore, the numbers of layers used in LeNet [16], AlexNet [8], and VGG16 [17] are from 5 to 6, and then to 16.

Despite the fact that neural networks with more training parameters can learn more complex functions, with increasing the number of layers, several problems may appear, such as gradient vanishing and exploding, which lead to incapacity to fully exploit the parameters for training. To deal with these problems, ResNet [18] proposed a cross-layer connection feature extraction method, called residual block, to alleviate the problems of gradient vanishing and exploding. Additionally, DenseNet [19] uses fully connected layers, while GoogleNet [20–23] uses Inception architecture to propagate the residual value. The cross-layer connection architecture can usually alleviate the gradient problem. However, Veit et al. [24] and Wu et al. [25] indicated that in a residual network, most paths are shorter than expected, and only short paths can provide effective gradients during the training process. Therefore, the vanishing gradient problem cannot be solved in the residual network by preserving the gradient flow throughout the entire depth of the network.

Many research studies [25–28] show that shallow feed-forward networks can learn the complex functions and obtain the accuracy only achievable with DCNNs. Additionally, the shallow neural networks can be trained to obtain the performances achievable with DCNNs by using the total number of parameters similar to that used in the DCNNs.

Many existing well-known DCNN architectures with a certain depth extent may have the problems mentioned above, and require many parameters for the training process, which is quite time and energy-consuming. Therefore, getting good identification accuracy while maintaining lesser neural networks' depth is an important research trend in the future. The number of used parameters is far less in MSSCNN and MSPCNN than those in DCNNs. They apply multi-scale subnetworks to extract the features of different scales so as to effectively identify the rice leaf spots affected by various degrees of infection.

2. RELATED WORKS

In this section, popular CNNs and multi-scale CNNs architectures will be briefly introduced. Some CNN based rice leaf spots identification methods will also be reviewed.

2.1 CNN Architectures

There are spatial relationships between the adjacent pixels in an image. CNNs [8, 16–23] mainly use trainable weight kernels to extract the features of an image. Generally, the convolution operation in a layer of CNN is described as:

$$I^l(x, y) = f \left(\sum_{i=-\lfloor S_{kx}/2 \rfloor}^{\lfloor S_{kx}/2 \rfloor} \sum_{j=-\lfloor S_{ky}/2 \rfloor}^{\lfloor S_{ky}/2 \rfloor} I^{l-1}(x+i, y+j) \times K^{l-1}(i, j) \right), \quad (1)$$

where $I^l(x, y)$ is the intensity of the pixel locating at (x, y) in the l th layer feature map; $K^{l-1}(i, j)$ is the weight at coordinates (i, j) in the $(l-1)$ -th layer convolution kernel with size $S_{kx} \times S_{ky}$; $f(\cdot)$ is a non-linear activation function.

The pooling layer aims to reduce feature dimensions. Without loss of generality, all features in a patch of the feature map are summarized into one to reduce dimensions. The max pooling and average pooling are the two most widely used operations, wherein the max pooling operation takes the maximum value of all the pixels in the patch, while the average pooling operation takes the average of all the pixels in the patch.

After the convolutional layers and the pooling layers, the extracting features will be connected through a full connection layer. When given the input images and their corresponding categories (labels), the learning process of CNN is responsible for training the fittest weights.

Back propagation learning is one of the most commonly used methods to optimize the weights of training networks. It exploits the calculated gradients to correct the weights of training networks, where the gradient is calculated as follows:

$$\Delta w_i = \frac{\partial E}{\partial w_i}, \quad (2)$$

where w_i is the weight to be corrected; E is the error value, namely, the difference between the current output of the network and the actual ground truth. Through the formula, the chain rule is used to propagate the error back to the weights, and then the gradient of the weight can be recalculated. The chain rule of differentiation is a composite function. When increasing the number of layers, the gradient terms to multiply will increase on the function. For a very deep CNN, if all the gradient terms are in the range $(0, 1)$, the gradient will decrease. We call this phenomenon a vanishing gradient problem. On the contrary, if all the gradient terms are greater than 1, it becomes an exploding gradient problem. A simple way to deal with this problem is to reduce the number of layers.

He et al. [18] provided a neural network with 20 layers which provides a better performance than the other network with 56 layers. Even if it is deepened by adding more hidden layers, the obtained performances for both training data and test data are even worse. Obviously, it is not the phenomenon of overfitting. Ref. [18] hence proposed a residual block architecture to reduce overfitting phenomenon. Nevertheless, [24, 25] believes that the effect

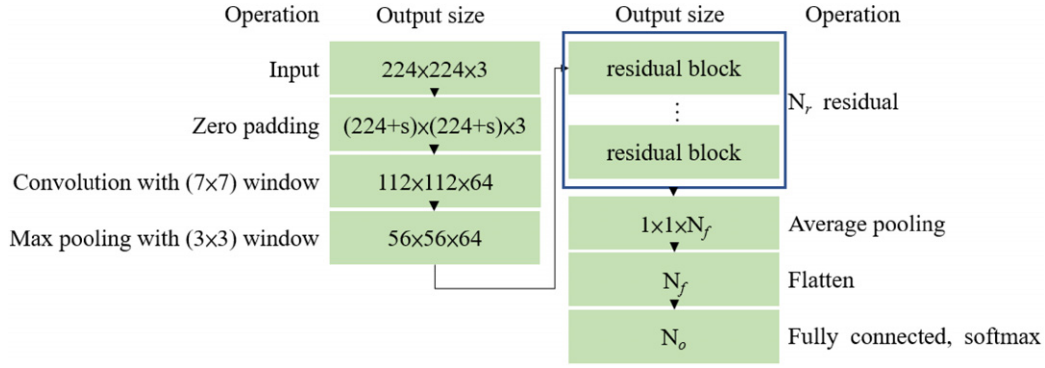


Figure 1. Residual Network architecture.

of residual block architecture is still limited, especially when given too many layers.

Ref. [18] adds residual blocks to the network for building a deeper neural network, which is called ResNet. It feeds the activation of a network layer to the subsequent network layers. The residual block is calculated as follows:

$$X_l = X_{l-1} + \sum_{i=0} F(X_i^{l-1}, W_i^{l-1}), \quad (3)$$

where X_l is a feature map in l th layer, and W_i^{l-1} is the given kernel's weights in the i th channel of the $(l-1)$ -th layer, and $F(\cdot)$ the convolution operation and the nonlinear activation function. Figure 1 shows the ResNet architecture proposed by Ref. [18], where it first goes through a convolution layer and a maximum pooling layer followed by N_r residual block layers, then an average pooling layer and a flatten layer are connected to convert the extracted feature maps into an N_f -dimensional vector, and a fully connected layer and Softmax function are finally used to output N_o neurons.

2.2 Multi-Feature Fusion CNNs

Despite that CCNs are successful in image processing tasks, single model-based CNNs cannot effectively extract the ROI (regions of interest) from images, especially for the images with large variations in object sizes [29, 30]. Small object size, multiple object scales, and complicated image background will make image classification more difficult. Different sensors can provide different perception information on the same context. Studies [31, 32] have confirmed that the feature fusion strategy of different convolutional layer feature maps can effectively solve these problems. Multimodal fusion uses different models to extract different perception features and combine them to improve the performances obtained by using only one modality.

There are three types of multimodal fusion architectures: early fusion, late fusion, and intermediate fusion [31, 32]. An early fusion architecture [33, 34] integrates multiple data before conducting the analysis. It will face a challenge when the sampling rates between modalities are different, or some sources are discrete while the others are continuous. A late fusion architecture [35–37] applies some independent

models to extract different features, and fuses them at a decision-making stage. Generally, a late fusion architecture is far simpler than the early fusion architecture, especially when the data sampling rate, data dimensionality, and unit of measurement are extremely varied from each other. However, the late fusion architecture needs more computational time because of the higher number of classifiers to be trained. Intermediate fusion architecture [38] is built on a deep neural network. It is the most flexible architecture in which the features can be fused at different stages of model training. Intermediate fusion based CNN fuses different modalities into a single hidden layer so that the model can learn a joint representation of many modalities; hence, it can greatly improve performance. In this research, MSSCNN is hence proposed, which is an intermediate fusion architecture.

DenseNet [19] and ResNet [18] are intermediate fusion architectures. ResNeXt [39] is an improved version of ResNet and it performs a set of transformations with the same topology, the outputs of which are aggregated by summation. The architecture can be extended to any large number of transformations without specialized designs. In the early stage of disease, rice leaves often contain many tiny lesions. In order to highlight these tiny lesions and avoid the influence of other symptoms, MSPCNN adopts a late fusion architecture.

2.3 Image Processing Based Rice Leaf Spots Identification

Phadikar et al. [40] first applied the Fermi energy based segmentation method to extract the infected portion from an image. The symptoms of diseases were then characterized and extracted based on the color, shape and spatial features of the infected portion. To reduce the complexity of identifying the rice leaves affected by rice leaf blast, rice sheath rot, or bacterial leaf blight, rough set theory is used to select the important features to minimize the loss of information. Mohan et al. [41] has utilized Haar-like features to detect the rectangular blocks where the infected spots are located, and then SIFT features are input to SVM and KNN respectively for identifying whether the rice leaves are infected with helminthosporium leaf blight, rice leaf blast or bacterial leaf blight. Abdullah et al. [42] has photographed rice leaves at a fixed distance, whose shape and color features were then



Figure 2. Diseased rice leaf images.

used for diagnosing disease symptoms. Xiao et al. [43] has extracted six color features, ten morphological features, and five texture features from each harvested lesion image, the principal component analysis (PCA) method was then used to map the 21 features into 6 comprehensive features, and then a back propagation (BP) neural network model was used to identify rice blast.

To improve the performances of the traditional image matching methods listed above, Liang et al. [44] has applied DCNN, which can extract higher level features, to recognize the rice blast disease. They used LeNet architecture to diagnose rice blast disease based on the rice leaf image, wherein every rice leaf image was divided into patches, each with 128×128 pixels, and then each patch was used to identify the rice blast disease by an approximate LeNet-5 architecture [16], which adopts SVM as the classifier. Hassan et al. [45] has also proposed a CNN architecture for plant disease diagnosis which uses depthwise separable convolution to improve the inception architecture. For diagnosing the nutritional deficiency of rice plants, Sharma et al. [46] has combined such classifiers as InceptionResNetV2, Xception, DenseNet201, and VGG19 to extract different features and fuse them into the average strategy.

3. MULTISCALE CNN-BASED RICE LEAF DISEASE IDENTIFICATION

The purpose of this research is to develop a multiscale convolutional neural network to identify the rice leaf infected with rice leaf blast, helminthosporium leaf blight, and bacterial leaf blight. As mentioned in Section 2, the design principle of the network is to develop a shallower network with as few parameters as possible. In this research, MSSCNN is hence proposed to increase the number of kernels for getting better identification effects, while MSPCNN is provided to extract more complete features of diseased rice leaves.

3.1 Experimental Samples

A set of field-taken images of common rice leaves affected by rice blast, helminthosporium leaf blight, and bacterial leaf blight from the areas south of Changhua County, Taiwan, was selected and labelled according to the ground

Table I. Number of diseased rice leaf images.

	No. of images	No. of image Infected blocks
Bacterial leaf blight	259	3207
Rice leaf blast	237	6417
Helminthosporium leaf blight	330	2926

Table II. Number of spot image blocks.

	No. of image blocks
Infected	12550
Healthy	29961

truth given by phytopathologists for the purpose of this research. Grand total of 826 rice leaf images, containing both infected and healthy examples, was selected for the study; the corresponding quantities are shown in Table I. Figure 2 shows some images of rice leaf spots, where the first row shows the images of bacterial leaf blight, characterized by the spot's infecting water pores of rice leaf (at leaf tip, leaf edge, leaf hole) and lighter color; the second row shows rice blast, characterized by the spot's spindle shape, smaller brown halo and darker color; the third row shows helminthosporium leaf blight, characterized by the spot's round shape and brown color. Therefore, the spot types can be identified by the foregoing disease characteristics shown in the rice leaf images.

In this research, each image was divided into several image blocks, each with size 200×100 pixels, as shown in Figure 3(a). According to the disease characteristics diagnosed by phytopathologists, these image blocks were then labeled as four types: healthy, rice leaf blast, helminthosporium leaf blight, and bacterial leaf blight. The number of healthy image blocks is far greater than that of infected image blocks, as shown in Table II. Therefore, the infected image blocks were further subject to horizontal and vertical flipping to increase the number of infected image blocks, as shown in Fig. 3(b). In addition, to make the model more generalizable, two-stage data augmentation [47, 48] will also be used to increase the size of training and testing data. The

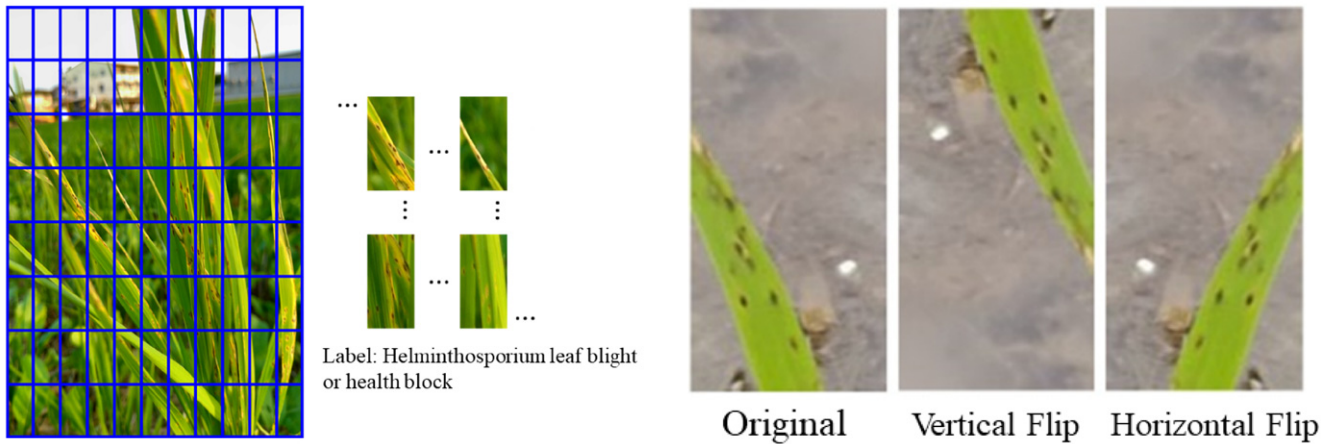


Figure 3. (a) Divided image blocks labeled as the corresponding types (helminthosporium leaf blight as an example). (b) Data Augmentation.

two-stage data augmentation will be described in detail in the subsequent section Experiment Results. MSSCNN and MSPCNN will then be used to identify the rice leaves infected with rice leaf blast, helminthosporium leaf blight, or bacterial leaf blight and conclude the research.

3.2 3-Conv-Layer ResNet

Many well-known CNNs have been proposed, such as LeNet [16], ResNet [18], and Inception-v4 [23], which use tens of thousands to tens of millions of parameters. For example, ResNet increases its depth from 18 layers to 34 layers, and even to 50 layers, and the number of parameters is also increased from 12.59M to 22.68M, and then to 23.59M. Similar situations are also found in DenseNet [19] and Inception-v4 [23].

In order to use a shallower network and as few parameters as possible, an identification model with 3-Conv-Layer ResNet, which uses only three convolution layers, is hence proposed for this research and is illustrated in Figure 4. The model also uses only one residual block layer to save computing time:

Input layer: a layer where bilinear interpolation is applied to convert each input color image into one with 112×56 pixels in size, containing 3 color components red, green, and blue.

Zero padding layer: a layer where S_k rows and S_k columns of zeros are added at the top, bottom, left and right sides of the input image from the upper layer and $S_k = (k - 1)/2$, meaning that the size of convolution kernel to be used in next layer contains $k \times k$ pixels in size.

$k \times k$ -Convolution layer: a layer where 32 kernels of $k \times k$ parameters in size are given; for each kernel, convolution operation is performed against the R , G , B channel images respectively and the 3 kernel maps subject to convolution operation are then added to form a kernel map.

Max pooling layer: a layer where a window of 3×3 pixels in size is used to perform a max pooling operation with stride $s = 2$ to convert the input data to a $56 \times 28 \times 32$ kernel map.

Convolution (3×3)-1 layer: a layer where 2 kernels of 3×3 pixels in size are given to perform convolution operation to convert the input image to a $28 \times 14 \times 64$ kernel map, where stride s is set to 2.

Convolution (3×3)-2 layer: a layer where one kernel of 3×3 pixels in size is given to perform convolution operation to convert the input image to a $28 \times 14 \times 64$ kernel map, where stride s is set to 1.

Add layer: a layer where the input data from Convolution (3×3)-1 layer and Convolution (3×3)-2 layer are added to generate a $28 \times 14 \times 64$ kernel map.

Average pooling (28×14) layer: a layer where average pooling operation is used to convert each 28×14 kernel map to a 1×1 kernel map; since there are 64 kernel maps, a $1 \times 1 \times 64$ kernel map is therefore generated at the output end.

Flatten layer: a layer where a $1 \times 1 \times 64$ kernel map is flattened into a one-dimensional array of 64 features.

Fully connected ReLU layer: a layer where ReLU function is used as the activation function to convert 64 input values to 32 outputs through the fully connected neural network.

Fully connected Softmax layer: a layer where Softmax function is used as the activation function to convert 32 input values to 4 outputs through the fully connected neural network, wherein the output value 0 represents the category of a healthy leaf image block; 1 is the category of a rice blast infected leaf image block; 2 is the category of a helminthosporium leaf blight infected leaf image block and 3 is the category of a bacterial leaf blight infected leaf image block.

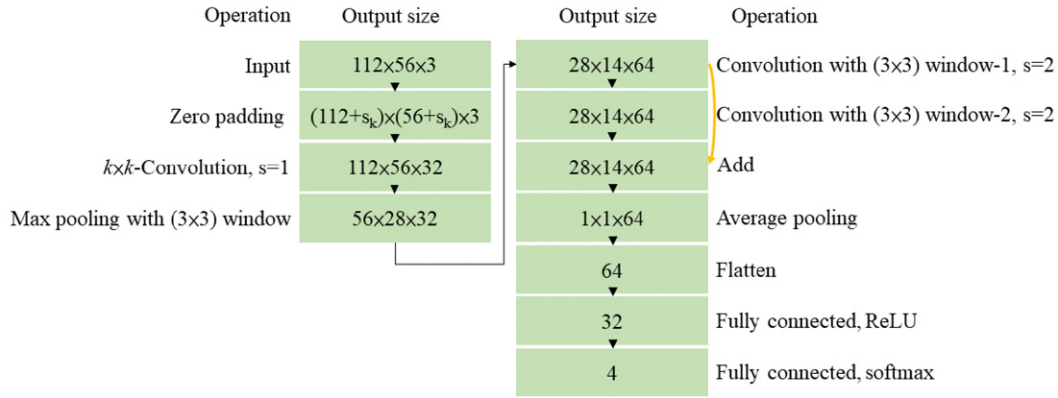


Figure 4. Architecture of 3-Conv-Layer ResNet.

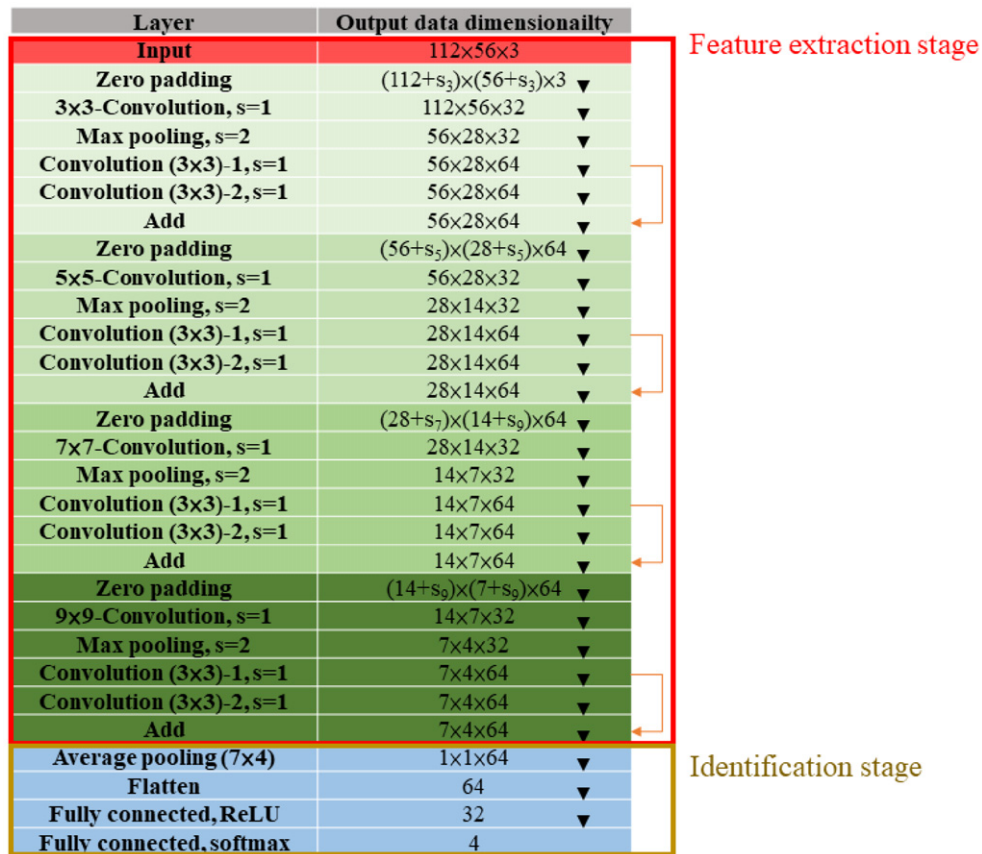


Figure 5. Architecture of MSSCNN.

3.3 Multi-Scale Serial Convolution Neural Network (MSSCNN)

A multi-scale serial convolution neural network (MSSCNN) is also proposed for this research. MSSCNN is an intermediate fusion architecture in which the features of different scales are fused at distinct stages of model training. The architecture of MSSCNN is roughly divided into feature extraction and identification stages. The feature extraction stage (as shown in the red box on Figure 5) is to extract the image features of different scales through convolution operations and average pooling operations, while the identification stage

(as shown in the yellow box on Fig. 5) is to perform category identification via the extracted image features incorporated with fully connected layer neural network.

Fig. 5 displays the architecture of this MSSCNN, in which bilinear interpolation is first applied to transform each input image into a one of 112×56 pixels in size. In order to extract the features of rice leaf spots in different sizes, MSSCNN consecutively connects four 3-Conv-Layer ResNets with four $k \times k$ -Conv layers, where $k = 5, 7, 9$, and 11 respectively; Each combination of 3-Conv-Layer ResNets with $k \times k$ -Conv layers hereby contain: Zero padding layer,



Figure 6. Small leaf spots in the left figure, and large ones in the right figure.

3×3 -Convolution layer, $k \times k$ -Convolution layer, Max pooling layer, Convolution (3×3)-1-layer, Convolution (3×3)-2 layer and Add layer. The architecture before the Add layer is exactly the same as in Fig. 4. The features of one scale will be extracted in a consecutive stage from an output of the previous stage's feature extraction of another scale. There is no need to turn the output features into the one-dimensional ones, so MSSCNN hence removes the average pooling layer and its subsequent layers as depicted in Fig. 4.

After the final 3-Conv-Layer ResNets and 11×11 -Conv layer combination, an average pooling (7×4) layer is used to convert each 7×4 kernel map to a 1×1 kernel map; since there are 64 kernel maps, a $1 \times 1 \times 64$ kernel map is generated as the output. Finally, the flatten layer, fully connected ReLU layer, and fully connected Softmax layer, exactly the same as their corresponding ones shown in Fig. 4, are applied to identify the input image block as a healthy rice leaf or a diseased rice leaf infected with rice leaf blast, helminthosporium leaf blight, or bacterial leaf blight based on the output feature maps of the average pooling (7×4) layer.

3.4 Multi-Scale Parallel Convolution Neural Network (MSPCNN)

As shown in Figure 6, the sizes of rice leaf spots vary with severity of disease. The probability of large spots appearing on the rice leaf affected by a more severe disease is usually higher. In this research, 3-Conv-Layer ResNet with different convolution kernel sizes is proposed to extract the features from rice leaf spots of different sizes. That is to say, the architectures of 3-Conv-Layer ResNet adopted exactly the same as those shown in Fig. 4, with the only difference in the k of $k \times k$ -Convolution, where in $s = 1$ layer, k is set to 5, 7, 9, and 11 respectively, and these four scale choices come from subsequent ablation experiments. Table III shows the experimental results obtained by the 3-Conv-Layer ResNet with different k . The obtained identification accuracy by four basic 3-Conv-Layer ResNet with $k = 3, 5, 7, 9$, and 11 are 90.92% (with $k = 3$), 91.73% (with $k = 5$), 90.41%

Table III. Experimental results obtained by 3-Conv-Layer ResNet with different k .

k	Accuracy
3	90.92%
5	91.73%
7	90.41%
9	91.45%
11	91.01%

(with $k = 7$), 91.45% (with $k = 9$), and 91.01% (with $k = 11$) respectively.

MSPCNN combines four different subnetworks to fuse the features of various scales. Similarly, MSPCNN consists of feature extraction and identification stages. The feature extraction stage (as shown in the red box on Figure 7) is to extract the image features of different scales through four 3-Conv-Layer ResNets with different k , while the identification stage (as shown in the yellow box on Fig. 7) is responsible for the category identification via the extracted image features incorporated with fully connected layer neural network.

In addition, batch normalization [21] is performed at each convolution layer so that each input datum is converted to the standard normal distribution. Stochastic gradient descent is used for the training model, and 300 epochs are performed in feature extraction stage and 10 epochs in classification identification stage, wherein the learning rate is set to 0.001 and momentum to 0.9, and cross-entropy is used as the loss function.

4. EXPERIMENT RESULTS

4.1 Comparison with Existing Well-Known Model

This subsection is dedicated to experimental comparison of the performances of MSSCNN, MSPCNN and several existing networks. Table IV demonstrates the experimental results. The architecture of LeNet-5 is straightforward and simple to understand. In ResNets, the outputs of

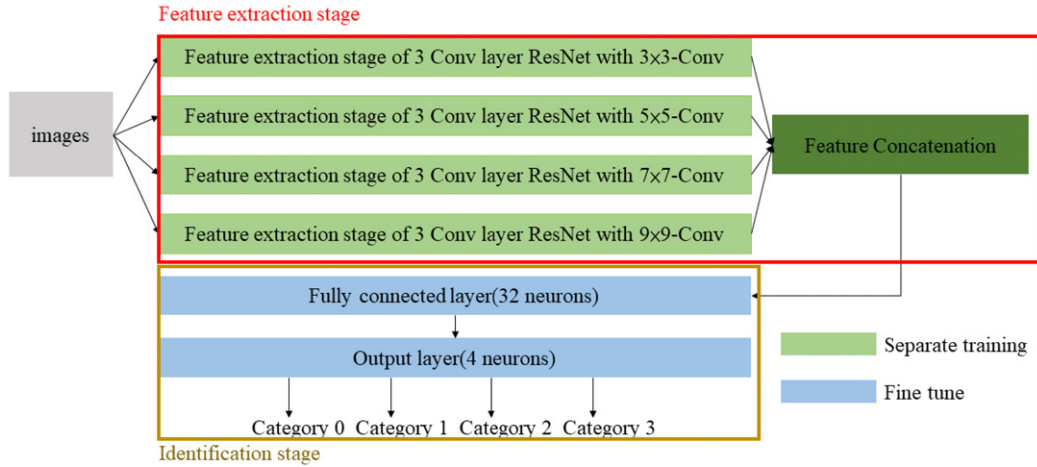


Figure 7. Architecture of MSPCNN.

Table IV. Experimental results of various CNNs.

Method	Accuracy	No. of parameters	Training time	% of non-zero weights
LeNet -5 [16, 44] model-1	81.66%	≈65.65K	≈6900 s (batchsize = 64)	90.19%
LeNet -5 [16, 44] model-2	82.79%	≈55.63K	≈6300 s (batchsize = 64)	87.35%
ResNet-18 [18]	91.04%	≈12.56M	≈49200 s (batchsize = 64)	65.20%
ResNet-34 [18]	91.15%	≈22.68M	≈79200 s (batchsize = 64)	66.25%
ResNet-50 [18]	89.49%	≈23.59M	≈133200 s (batchsize = 64)	73.78%
ResNeXt-50 [39]	91.06%	≈23.59M	≈191356 s (batchsize = 64)	83.85%
DenseNet-121 [19] $k = 32$	91.41%	≈7.04M	≈186600 s (batchsize = 32)	86.73%
DenseNet-169 [19] $k = 32$	91.48%	≈12.64M	≈238200 s (batchsize = 32)	85.39%
DenseNet-201 [19] $k = 32$	91.45%	≈18.32M	≈365700 s (batchsize = 16)	84.14%
GoogLeNet with Inception-v4 [23]	89.56%	≈41.18M	≈375000 s (batchsize = 32)	79.55%
MSSCNN	91.64%	≈0.82M	≈12780 s (batchsize = 64)	81.37%
MSPCNN	92.52%	≈0.33M	≈7700s+3400 s (batchsize = 64)	87.59%

the preceding layers can reach the succeeding layers with minimal obstructions. Based on LeNet-5, Liang et al. [44] has proposed LeNet-5-model-1 and LeNet-5-model-2. Despite using less parameters and spending less training time (each model was also trained on Nvidia GeForce GTX 1080Ti), LeNet-5-model-1 and LeNet-5-model-2 have the identification accuracy of only 81.66% and 82.79% respectively.

Cross layer connection is the main characteristic of ResNet, in which the output is passed across layers and added to the input of convolution. This connection can fuse feature maps of different granularity. The experimental results show that ResNet with 18, 34, and 50 convolution layers [18] provide the identification accuracies 91.04%, 91.15%, and 89.49% respectively. However, increasing number of convolution layers does not increase accuracy and causes an increase in the number of parameters, with the number of parameters increasing from 12.56M to 22.68M and 23.59M respectively.

ResNeXt applies a much simpler design than all Inception models. In ResNeXt, the blocks generating feature maps of the same dimensionality share the same set of hyperparameters. Furthermore, if all the feature maps are downsampled by a factor of 2, the width of the block is multiplied by a factor of 2. In this experiment, a ResNeXt-50, constructed by a template with cardinality = 32 and bottleneck width = 4d, is applied to identify the diseased rice leaves. Table IV illustrates that ResNeXt can obtain an identification accuracy of 91.06% and use almost the same number of parameters, but spend more training time than ResNet-50. The ResNet-based architectures give almost the same performances as the 3-layer and 18-layer or even 50-layer networks. The same phenomenon also occurs in other cross-layer connection networks: DenseNet with 121, 169, and 201 convolution layers [19]. In the experiment, DenseNet with 121, 169, and 201 convolution layers provide the identification accuracies 91.41%, 91.48%, and 91.45% respectively. Likewise, the identification accuracy does not

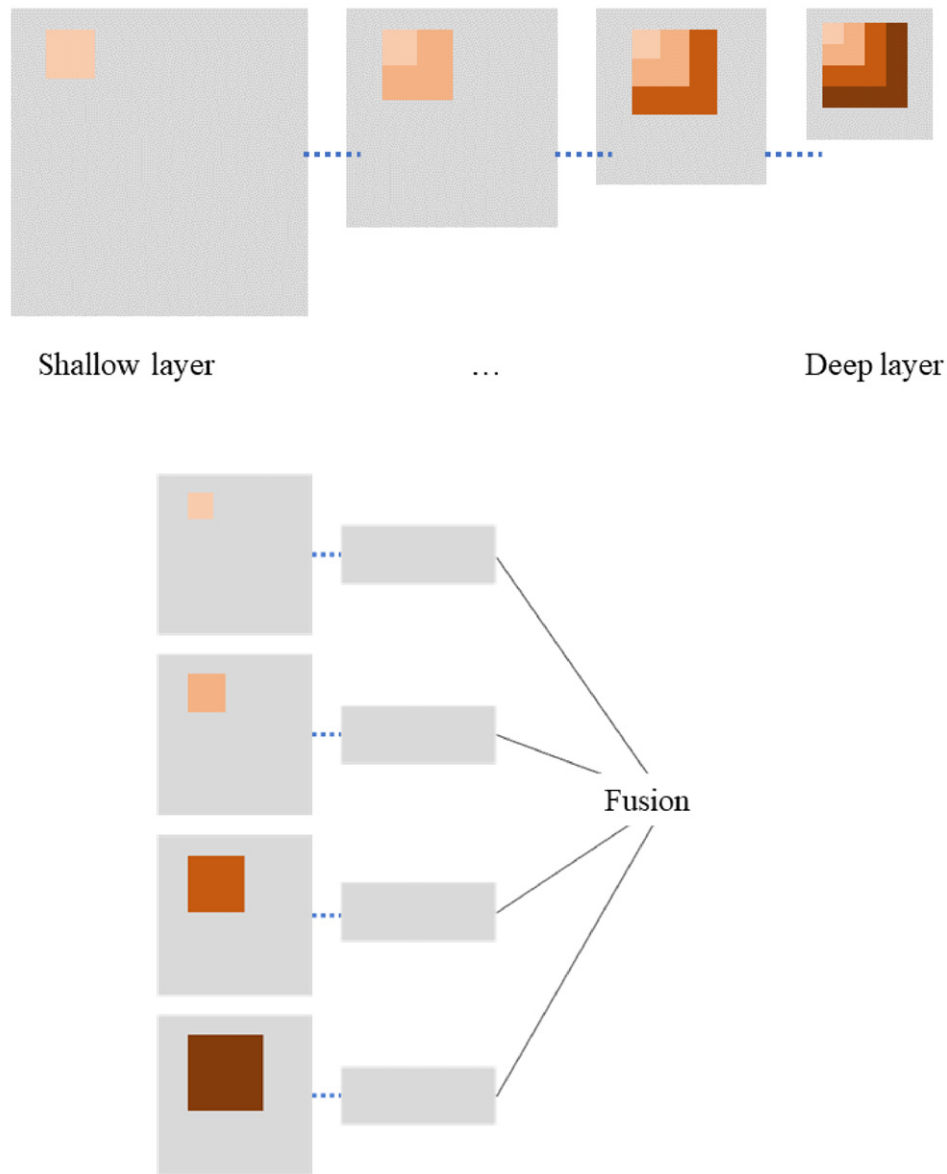


Figure 8. (a) Features extraction behavior of MSSCNN and (b) Features extraction behavior of MSPCNN.

increase with increase in the number of convolution layers, either.

Inception-v4 is a DCNN architecture simplifying previous iterations of the Inception family, adding stem layers and using more Inception modules than Inception-v3, with a more uniform simplified architecture. In this experiment, a large-scale network Inception-v4 [23] is taken to detect the diseased rice leaves. It uses a cross-layer connection strategy and a large number of parameters, but its performance is still lower than the performance of the two networks listed above.

From Table IV, one can quickly observe that increasing the number of layers of CNN does not necessarily lead to a higher identification accuracy. As the number of layers increases, the number of parameters increases as well. The strategy provided in this research hence employs multi-scale features to enrich the features instead of increasing the

number of layers. MSPCNN hence uses a shallower network structure. From Table IV, MSPCNN uses much less number of parameters but obtains a higher accuracy since it offers a bigger variety of features.

In a convolutional layer, each neuron gets input from the previous layer. A CNN consists of multiple layers of artificial neurons. The first layer is usually to extract basic features, i.e., horizontal or diagonal edges. This output is then passed to the next layer for detecting more complex features, i.e., corners or combined edges. As it goes deeper into the network, more complex features can be recognized, i.e., objects, faces, etc. Therefore, each neuron takes input from a larger area of feature maps than that of the previous layers.

Although MSSCNN adopts cross-layer connections to solve the feature extraction problem of smaller spots, the features of tiny lesions on diseased rice leaves may still

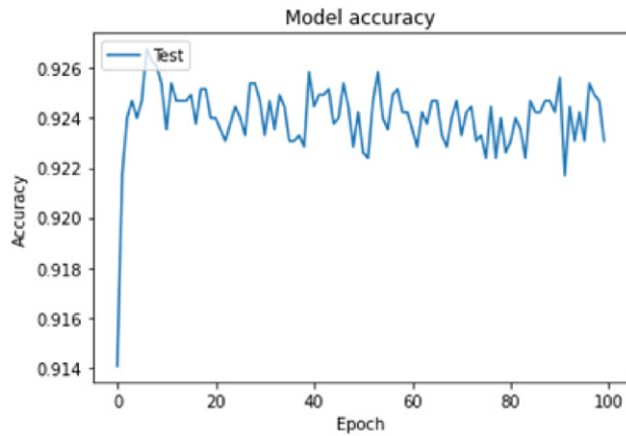


Figure 9. The convergence of MSPCNN.

be destroyed. Figure 8 illustrates the difference between the feature extraction behaviors on shallow layers and deep layers in MSSCNN and MSPCNN. MSSCNN adopts a direct concatenation architecture which provides tighter connections than MSPCNN, so it employs a larger number of parameters. Table IV shows that MSSCNN uses about 0.82M parameters, which is more than that of MSPCNN. Additionally, MSSCNN offers an accuracy of 91.64% in identifying the diseased rice leaves, slightly inferior to MSPCNN. To investigate the convergence of MSPCNN by experiment, in the experiment, MSPCNN is run for 100 epochs in training stage, Figure 9 illustrates the obtained accuracy in each epoch.

When the computations performed by DNNs involve a multiplication by a zero weight, a large fraction of the computations will be intrinsically ineffectual [49]. Table IV shows that MSPCNN provides a higher ratio of non-zero weights than others except LeNet-5 model-1. Since LeNet-5 model-1, LeNet-5 model-2, MSSCNN, and MSPCNN use lower rates of zero weights, they consume less parameters and training time. Gradient vanishing may result in more approximate zero weights. Although DenseNet and Inception provide acceptable non-zero weights rates, they use fairly deep networks which may create approximate zero weights.

In this paper, the performances of the CNN architectures with multi-feature fusion proposed by Hassan et al. [45] and Sharma et al. [46] are also illustrated by experiments in identifying the diseased rice leaf infected with rice leaf blast, helminthosporium leaf blight, and bacterial leaf blight. Table V demonstrated the experimental results where MSPCNN is superior to the architectures provided by Hassan et al. and by Sharma et al. in identification accuracy and the number of parameters consumed. The reason is that Hassans' architecture uses early fusion which may cause losing many tiny features while Sharmas' architecture adopts a deep CNN structure which has more parameters.

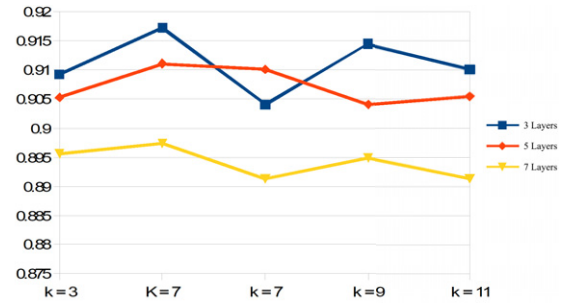
Figure 10. The performances of three-, five-, and seven-layer RseNets with different k .

Table V. Comparison with multi-feature fusion CNNs.

Method	Accuracy	No. of parameters
The method of Hassan et al. [45]	91.22%	$\approx 0.42M$
The method of Sharma et al. [46]	92.01%	$\approx 18.32M + 54.34M$
MSPCNN	92.52%	$\approx 0.33M$

4.2 Comparison of Number of Layers

MSPCNN adopts three-layer RseNets with k of 5, 7, 9, and 11 respectively. Next experiment is to explore the performances of three-, five-, and seven-layer RseNets respectively. Moreover, all k s are set to all 5s, all 7s, all 9s, and all 11s, respectively. Figure 10 shows the experimental results which indicate that increasing the number of layers may not necessarily be effective in improving performance. The experimental results also show that the three-layer RseNets architecture can achieve better results. This is why MSPCNN uses three layers of RseNets.

4.3 Ablation Experiment

The fourth experiment in this research is to detect the most suitable composition of subnetworks with different scales for MSPCNN. In order to create the most suitable combination of subnetworks of different scales, the third experiment probes the identification accuracy of six combinations of subnetworks with two different scales, and four combinations of subnetworks with three different scales. In addition, it also investigates the performances of the combination of subnetworks with four different scales and the combination of subnetworks with five different scales. Table VI demonstrates the experimental results. Apparently, the combination of subnetworks with four different scales ($k = 3, 5, \dots, 11$) can provide the best recognition accuracy.

4.4 Two-Stage Data Augmentation

Data augmentation has been effectively used to improve the robustness of DCNNs, especially when the training data is limited. Zheng et al. [47, 48] adopted a two-stage data augmentation strategy to enhance the generalization of a model. A two-stage data augmentation strategy is also implemented in this research to generate more varied image samples in order to improve the identification accuracy. In

Table VI. Ablation experiment for number of scale subnetworks.

Scales combination (k)	Accuracy	Scales combination (k)	Accuracy
3 + 5	91.66%	3 + 7 + 9	92.01%
3 + 7	91.41%	3 + 7 + 11	92.17%
3 + 9	91.31%	3 + 9 + 11	92.22%
3 + 11	91.68%	5 + 7 + 9	92.22%
5 + 7	92.01%	5 + 7 + 11	92.19%
5 + 9	91.91%	5 + 9 + 11	92.10%
5 + 11	91.68%	7 + 9 + 11	91.75%
7 + 9	92.05%	3 + 5 + 7 + 9	92.26%
7 + 11	91.27%	3 + 5 + 7 + 11	91.96%
9 + 11	92.01%	3 + 5 + 9 + 11	92.08%
3 + 5 + 7	91.38%	3 + 7 + 9 + 11	92.17%
3 + 5 + 9	91.94%	5 + 7 + 9 + 11	92.52%
3 + 5 + 11	92.15%	3 + 5 + 7 + 9 + 11	92.19%

the last experiment, for each image I_o (may be a training image or a testing image), horizontal flip augmentation operation is performed to produce a variant image I_h , and vertical flip augmentation operation is employed to create another variant image I_v . In the training stage, I_o , I_h , and I_v of all training images are used to train the MSPCNN model.

In the testing stage, for a given testing image I_o , MSPCNN uses three modes respectively to identify which diseased rice leaves I_o , I_h , and I_v are. If the confidence of identifying either I_h or I_v is less than a threshold $Th_c (= 0.5)$, MSPCNN will give up the results obtained in identifying I_h and I_v , and only agree with the result of detecting the I_o . If both confidence values of detecting I_h and I_v are larger than or equal to Th_c , MSPCNN will regard this image as the diseased rice leaf which most detecting modes support. The experimental results show that, combined with this strategy, the identification accuracy of MSPCNN is improved from 92.52% to 93.71%. The experimental results also show that if combined with the two-stage data augmentation strategy provided by Zheng et al. [47, 48], the identification accuracy of MSPCNN will reduce from 92.52% to 92.12%.

5. CONCLUSIONS

Rice leaf blast, helminthosporium leaf blight, and bacterial leaf blight are the three most common rice diseases that seriously damage the rice yield. Consequently, it is necessary to develop an automated diagnosis system to identify these three rice diseases automatically. Normally, the deeper the CNN is, the more likely it is to fit a large number of diverse data features. However, the experimental results depict that when the number of parameters of a CNN is increased to a certain extent, the CNN's accuracy does not actually increase with the number of parameters, the required amount of computations, however, increases greatly. Two CNN architectures, MSPCNN and MSSCNN, which are used to identify rice leaf spots affected by rice leaf blast, helminthosporium leaf blight, or bacterial leaf blight, are

therefore proposed in this research. MSPCNN and MSSCNN use multi-scale subnetworks to extract the spot features of various rice leaf scales. Experimental results show that, compared with other CNNs, MSPCNN can provide the best identification accuracy, and use the least amount of parameters and training time. In this research, two-stage data augmentation is also applied to further improve the identification accuracy to 93.71%.

MSPCNN is a late fusion architecture and the features from each scale subnetwork was extracted independently, so that the characteristics of the tiny spots on the diseased rice leaves can be preserved intact. Hence MSPCNN can achieve a better performance than MSSCNN. The ablation experimental results demonstrate that MSPCNN using 3-Conv-Layer ResNet with $k = 5, 7, 9$, and 11 shows performance, that is superior to others. Another advantage of the MSPCNN architecture is that it can be implemented in a parallel computing environment.

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